

AI-Based Factory Worker Safety Monitoring System

Domain: Industry Safety | Computer Vision | Multimodal AI | MLOps

Problem Description

Industrial workplaces such as factories, warehouses, and construction sites are highly prone to accidents and unsafe working conditions. Workers often forget to wear personal protective equipment (PPE) such as helmets, gloves, masks, and safety vests. Most industries still rely on manual supervision and traditional CCTV systems, which cannot continuously monitor every worker in real time. Existing systems mainly focus on a single safety aspect and fail to provide integrated monitoring of worker behavior, safety compliance, and emergency situations. As a result, critical warning signs such as PPE violations, worker falls, dangerous machine interaction, abnormal sounds, and safety complaints often go unnoticed until accidents occur. There is a need for a simple and intelligent AI-based monitoring system that can automatically detect unsafe worker activities and generate alerts before accidents happen.

Objective

The objective of this project is to develop a beginner-friendly multimodal AI-powered worker safety monitoring system using image, audio, text, and video data. The system aims to:

- Detect workers without helmets, masks, or gloves

- Identify unsafe movement and restricted area entry
- Detect worker falls and dangerous machine interaction
- Analyze emergency sounds such as screams and alarms
- Process safety complaints and incident reports
- Generate real-time safety alerts
- Improve industrial worker safety using AI
- Support continuous monitoring through an end-to-end MLOps pipeline

Proposed Solution

The proposed system integrates four multimodal data sources: **Video (CCTV Monitoring)**
Detects unsafe worker movement, worker falls, restricted zone entry, and unsafe machine interaction.

Audio (Factory Sound Monitoring)

Detects scream sounds, emergency alarms, abnormal machine sounds, and distress signals.

Image (Worker Safety Images)

Detects helmet detection, mask detection, gloves detection, and PPE compliance monitoring.

Text (Safety Reports & Complaints)

Analyzes worker complaints, incident reports, hazard descriptions, and safety feedback. Each modality is processed independently using dedicated AI models and fused together using a weighted fusion mechanism to generate a final worker safety risk score.

System Workflow

1. Data Collection – CCTV video feeds, worker safety images, factory audio signals, and safety reports.

2. Preprocessing – Image resizing, audio filtering, text cleaning, and video frame extraction.
3. Model Processing – Computer Vision, Audio Classification, and NLP models.
4. Fusion Layer – Combine modality outputs using weighted ensemble techniques.
5. Risk Classification – Safe, Low Risk, Moderate Risk, and High Risk.
6. Alert & Dashboard – Real-time alerts, monitoring dashboard, reports, and visualization.
7. MLOps Pipeline – Continuous monitoring, automated retraining, deployment, and performance tracking.

Expected Outcome

- Reduce workplace accidents
- Improve worker safety compliance
- Detect unsafe conditions early
- Send instant alerts during emergencies
- Help industries monitor workers automatically
- Provide a scalable multimodal AI-based safety solution

Innovation & Impact

- Integrates text, audio, image, and video in one system
- Combines industrial safety monitoring with AI automation
- Beginner-friendly real-world implementation
- Supports real-time worker safety monitoring
- Implements an end-to-end MLOps lifecycle
- Provides scalable smart factory safety architecture

Technologies Used

Image Processing: OpenCV, YOLO, TensorFlow, PyTorch, Pillow

Audio Processing: Librosa, PyAudio, SoundFile

Text / NLP: BERT, NLTK, Scikit-learn

Video Analysis: OpenCV, MediaPipe, YOLO

ML Frameworks: PyTorch, Scikit-learn, NumPy, Pandas

App & Dashboard: Streamlit, Flask, SQLite, Plotly

Datasets

Image Datasets: PPE Detection Dataset, Helmet Detection Dataset, Roboflow PPE Dataset

Video Datasets: Worker Safety Surveillance Dataset, Industrial Activity Dataset

Audio Datasets: UrbanSound8K, ESC-50 Dataset

Text Datasets: OSHA Incident Reports, Workplace Safety Complaint Dataset

Conclusion

This project addresses an important industrial safety challenge by proposing a multimodal AI-driven worker monitoring system. By combining text, audio, image, and video analysis with MLOps principles, the system can detect unsafe worker activities, monitor safety compliance, and provide real-time alerts to prevent workplace accidents and improve industrial safety.